

L1 & L2 Processing Of Counterfactual Conditionals: Evidence From fMRI (Registered Report)

Abstract

The present fMRI study examines how native and non-native speakers process counterfactual conditionals with different modal verbs (root versus complex). While temporal sequencing in counterfactuals has been well-studied, critical gaps remain regarding (1) how modal complexity---root modals (e.g. volitional ‘would’) versus complex modals (e.g. hypothetical ‘could’)---impacts real-time processing, and (2) whether L1-L2 differences reflect distinct neurocognitive strategies for resolving modal ambiguities. We hypothesize that L1 Chinese speakers will show distinct activation patterns for root and complex modals, reflecting their syntactic-semantic versus conceptual demands. For L1-Chinese L2-English learners, we predict proficiency-dependent effects: intermediate learners will rely on prefrontal control regions for hypothetical modals, while advanced learners may show native-like processing only for simpler root modals. By combining parametric fMRI with cross-linguistic comparisons, this study provides crucial neurobiological evidence for how modal semantics interact with language experience, advancing theories of bilingual sentence processing. The findings will elucidate the neurocognitive costs of complex modal semantics.

Keywords: counterfactual conditional, fMRI, bilingual processing

1. Introduction

Counterfactual conditionals are semantic constructions that represent events contrary to reality or compare reality to a hypothetical alternative (Kulakova & Nieuwland, 2016). Theoretical linguistics research on counterfactual conditions with cross-linguistic temporal markers abounds in English and Chinese data. In English, the counterfactual sense is often cued by the flexible use of conditional connectives (e.g. ...if...), subjunctive mood, modal verbs (e.g. would, might), or explicit verbs (e.g. wish) (Ziegeler, 2000). Temporal indicators in Chinese (e.g. 明天 ‘tomorrow’, 去年

‘last year’, 当时 ‘at that time’) can serve as important cues to locate when a described event occurs and determine a hypothetical statement. In general, across languages, a higher level of counterfactuality tends to imply a lower level of factuality, and vice versa. Temporal indicators implying the past have been shown to enhance comprehension of counterfactuality compared with the condition without any temporal marker (Zhou, 2019). **Table 1** and **Table 2** show sample counterfactual conditionals.

Table 1. Sample English counterfactual conditional

<i>If I were you, I would try to fight for my rights.</i>

Table 2. Sample Chinese counterfactual conditional

早知今日，何必当初？
If only I had known the results today, why would I have done it in the first place?

The key discussion about counterfactual processing in psycho(-neuro)linguistics lies in the relevance between temporal sequence and neural responses of temporal indicators. Empirical evidence has demonstrated that human cognition expects the temporal event sequence to align with the chronological order of events in real life (e.g. Nieuwland, 2015). The Event-related Potential (ERP) results in *before*-guided sentences rather than *after*-guided sentences triggered sustained left frontal negativity, implying a higher cognitive load on readers’ working memory (Nieuwland, 2014). Similarly, neuroimaging findings corroborated that the *before*-guided conditionals elicited activations in the caudate nucleus and the left middle frontal gyrus more than the *after*-guided conditionals (Ye et al., 2012).

Concerning the correlation between temporal distance (not temporal sequence) and neural responses, it has been agreed that a longer temporal distance between the antecedent and the consequent struck a greater N400 (Ditman et al., 2008) or longer processing time than a shorter temporal distance (Speer & Zacks, 2005; Zwaan, 1996). To be more specific, the most recent neurolinguistic study clearly defines how temporal distance and chronological sequence as two factors would affect counterfactual processing (Liao et al., 2023) (see **Table 3**).

Table 3. Sample set of experimental stimuli

condition	sample sentence
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chronological, short	要不是/小王/昨天/没交论文, /他/此时/就不会/认真准备。 If only Wang had handed in his dissertation yesterday, he would not be preparing seriously now.
chronological, long	要不是/小王/去年/没交论文, /他/此时/就不会/认真准备。 If only Wang had handed in his dissertation last year, he would not be preparing seriously now.
reverse-chronological, short	要不是/小王/明天/要交论文, /他/此时/就不会/认真准备。 If only Wang wouldn't have to hand in his dissertation tomorrow, he would not be preparing seriously now.
reverse-chronological, long	要不是/小王/明年/要交论文, /他/此时/就不会/认真准备。 If only Wang wouldn't have to hand in his dissertation next year, he would not be preparing seriously now.

A chronological condition suggests that a subordinate clause denotes a time-frame earlier than the consequent/main clause, while a reverse one is the opposite. A short condition, representing a property of temporal distance, describes the distance between the temporal markers in two clauses. For instance, 明年 ‘next year’ and 此时 ‘now’ form a longer distance than 昨天 ‘yesterday’ with 此时 ‘now’.

Similar to Ditman et al.’s study (2008), Liao et al. (2023) employed temporal markers¹ (i.e. second VS hour VS year) in an *after*-guided clause to manipulate perceived temporal distance (e.g. *Sue pet the cat. After 1s/hour/year, the cat purred happily.*). However, a critical limitation of this approach is that the manipulated time-frames may conflict with world knowledge (e.g. a cat purring after one year is highly implausible), potentially conflating temporal distance with pragmatic plausibility. This confound may underlie the inconsistent neural findings between the two studies: while Ditman et al. (2008) observed a temporal shifting effect (increased N400 with longer distances), Liao et al. reported no such N400 effect for temporal indicators, suggesting that perceived difficulty may not linearly increase with temporal distance in decontextualized scenarios. Their significant findings were instead tied to chronological sequence, with reverse-chronological order eliciting

¹ It is important to note that the linguistic devices mentioned above (e.g. connectives, modals, temporal indicators) are not deterministic markers of counterfactuality. Rather, they are probabilistic cues that frequently lead to a counterfactual interpretation (e.g. I wish he were here. VS I wish he would come). Counterfactual interpretation may also hinge on contextual information. The present study operationalizes these cues as key experimental variables to investigate their relative weight and associated neural mechanisms during real-time sentence comprehension, following the established psycholinguistic approach of Liao et al. (2023).

characteristic ERP components (N400 and late positivity).

A few research gaps are worthy of being addressed by building on Liao et al.'s (2023) study and previous studies. First, a modal marker with varying types following the temporal indicator in the antecedent/subordinate clause has never been considered. Gibbs (1990) reveals that the acquisition order of modal verbs is closely tied to cognitive complexity: root modals that denote ability and permission are acquired earlier than epistemic modals that denote possibility and necessity, and counterfactual constructions precisely rely on the hypothetical meanings of epistemic modals (e.g. *could*, *might*). This developmental asymmetry suggests that the semantic complexity of modal verbs may impact cognitive load during real-time processing. For instance, modal verbs in counterfactuals (e.g. *If he could earn more money, he would buy a car.*) involve hypothetical reasoning about non-actual worlds, and Gibbs' finding that children's comprehension of such higher-order modals lags significantly behind root modals may imply that when adults process counterfactuals, the semantic features of modals (e.g. hypotheticality, non-actuality) may demand additional cognitive resources, thereby affecting processing efficiency.

Cournane (2020) further highlights the syntax-semantics complexity of modal verbs, arguing that their interpretation heavily depends on their syntactic position (e.g. root modals below tense, epistemic modals above tense) and their interaction with other grammatical categories (e.g. tense, aspect, negation). This insight supports the theoretical foundation for the processing mechanisms of counterfactual constructions: such constructions typically involve embedded past perfect tense (e.g. *had done*) and modal verbs (e.g. *could have done*). Cournane's proposed "modal meaning space" theory suggests that the interplay of force (e.g. possibility VS necessity) and flavor (e.g. root VS epistemic) of a modal may dynamically influence real-time syntactic parsing. For example, epistemic modals in counterfactuals may trigger deeper hierarchical processing because their semantics require integrating the logical relationship between non-actual worlds and counterfactual premises.

In sum, while these studies do not directly address processing, their insights into the semantic layering, syntactic constraints, and acquisition challenges of modal verbs provide a theoretical foundation for understanding their role in counterfactual processing: the semantic complexity and

syntactic specificity of modal verbs may significantly impact the real-time efficiency and neurocognitive mechanisms of counterfactual comprehension by increasing cognitive load or triggering ambiguity resolution.

Second, native and non-native real-time processing of counterfactual construction has yet to be compared by using neurolinguistic methodology, which requires more investigation. While prior work has established that modal verbs introduce semantic complexity and affect acquisitional trajectories (Cournane, 2020; Gibbs, 1990), how these markers modulate neural mechanisms differently across language backgrounds remains unknown. Comparing adult L1 and L2 processing via fMRI can reveal whether L2 learners exhibit altered neural signatures when resolving modal ambiguities in counterfactual construction, such as disambiguating root versus complex readings. This comparison is vital because L1 speakers may rely on automated syntax-semantics cues for modal interpretation, whereas L2 learners might depend more on explicit parsing, leading to distinct neural activations. The inclusion of modal markers further amplifies this contrast: their cross-linguistic variability in form-meaning mapping could exacerbate processing disparities between groups. Such insights would advance theories of language representation, clarifying whether modal-driven counterfactual processing involves shared neural substrates or diverges due to L2 proficiency, and shedding light on how semantic complexity interacts with language experience to shape neurocognitive architecture.

2. The Present Study

Motivated by the current research gaps, the present study addressed two inquiries:

- Do L1 Chinese speakers and L1-Chinese L2-English learners exhibit different profiles of processing counterfactual constructions with and without modal markers?
- How do proficiency levels modulate L1-Chinese L2-English learners' processing of counterfactual constructions with and without modal verbs?

While the aforementioned ERP research has been instrumental in delineating the temporal dynamics of counterfactual processing, the neural substrates and functional networks underpinning these

processes remain less explored. fMRI, with its superior spatial resolution, is uniquely positioned to address this gap. It allows for the precise localization of brain regions involved in the distinct cognitive components of counterfactual reasoning, such as syntactic unification, mental-state reasoning, and procedural learning, which may be differentially engaged by modal complexity and linguistic experience. Thus, the present study builds upon the established temporal profile from ERP research to provide a complementary, spatially precise map of the brain networks involved. This approach will elucidate how the neural circuits for language control and semantic integration, offering a systems-level understanding that bridges electrophysiological findings with their anatomical foundations.

Our preliminary hypotheses propose that L1 Chinese speakers and L1-Chinese L2-English learners will exhibit distinct neural profiles when processing counterfactual constructions with and without modal verbs, with these differences being modulated by both proficiency level and modal type. Critically, as reviewed, while developmental studies show that root modals (e.g. volitional *would*) are acquired earlier than complex modals (e.g. hypothetical *could* and epistemic *might*) in children, we predict adult processing will reveal a more nuanced pattern due to mature cognitive and linguistic systems. For the first research question, we hypothesize that L1 Chinese speakers will demonstrate relatively efficient processing of both root and complex modals in counterfactuals, as evidenced by reduced activation in prefrontal control regions compared to L2 learners. This prediction stems from adults' fully developed executive functions and extensive experience with modal semantics, allowing them to compensate for the inherent complexity of hypothetical modals through automatized processing routines (Nieuwland, 2015). However, subtle differences may emerge: root modals should elicit stronger engagement of the left inferior frontal gyrus (associated with syntactic unification), while complex modals may recruit the temporoparietal junction (linked to mental-state reasoning), consistent with their distinct semantic demands (Hagoort, 2013).

For L2 learners, we anticipate proficiency-dependent effects that mirror, but do not precisely replicate, the developmental trajectory. Intermediate learners (B2) are predicted to show heightened activation in the anterior cingulate cortex and bilateral prefrontal cortices when processing hypothetical modals, reflecting increased cognitive control demands during disambiguation of hypothetical meanings (Abutalebi & Green, 2016). In contrast, advanced learners (C1/C2) may

approach native-like patterns for root modals first, with complex modals continuing to require supplementary recruitment of the caudate nucleus, a structure implicated in procedural learning of complex form-meaning mappings (Ullman, 2001). This graded proficiency effect aligns with the prediction of the declarative/procedural model that L2 learners gradually proceduralize grammatical knowledge, with root modals (being conceptually simpler) transitioning to automatized processing earlier than complex modals.

The temporal organization of events (chronological VS reverse-chronological) is expected to interact with modal type, particularly for L2 learners. Reverse-chronological conditions containing complex modals may elicit the strongest activation in the inferior parietal lobule, a region sensitive to temporal sequencing conflicts (Ye et al., 2012), as learners struggle to simultaneously resolve non-linear time references and hypothetical modality. This effect may be attenuated in advanced learners, suggesting that proficiency mitigates combinatorial processing costs. Crucially, these predictions incorporate developmental evidence while accounting for adult neurocognitive adaptations: while children's incremental acquisition reflects emerging conceptual capacities, adults' fully matured executive systems and neural plasticity allow them to handle modal complexity through compensatory mechanisms, albeit with L1-L2 differences in the neural implementation of these processes.

3. Methodology

Experiment 1

Participants

We plan to recruit 40 neurotypical L1-Chinese L2-English learners with at least a university educational background (age range: 18-26), who will respond to both Chinese and English stimuli in this single experiment. A power analysis using GPower ($\alpha=0.05$, power=0.8) confirms that $n=30$ suffices to detect hypothesized activation differences in ROIs (left IFG, TPJ, and caudate nucleus). Interim analyses will not be conducted to avoid Type I error inflation. LexTALE is a validated quick test to offer a strong prediction of general English proficiency (Lemhöfer & Broersma, 2012), which will be used before the main experiment. The bilinguals, who began learning English at age 6,

will be categorized into intermediate and advanced groups based on a LexTALE cut-off score of 80%. The participants who scored below 60% (i.e. those not meeting the CEFR B2 threshold) will not proceed to the following experiment. To further characterize the sample and control for individual differences in language experience, all participants will complete a detailed language background questionnaire. This questionnaire will collect data on the age of acquisition, the length of formal instruction, current daily L2 usage, and immersion experience.

The participants will be excluded if head motion exceeds 3 mm translation or 3° rotation during fMRI scans, or if their comprehension question accuracy falls below 80%. Excluded data will not be replaced unless pre-planned quotas for group sizes (n=20 per proficiency level) are unmet, in which case additional participants will be recruited to maintain statistical power.

Materials

We start by investigating how the presence or absence of a root modal interacts with event chronology. The experimental stimuli will follow a 2×2 Latin square design. As reviewed, the short and long conditions had no effect on processing in Liao et al.’s (2023) study. Hence, the factor of temporal distance can be excluded from the present study. On the other hand, as the chronological and reverse chronological conditions triggered downstream neural activation in Liao et al.’s study (2023), it is worthwhile to reconsider the factor of chronology and examine its interaction with the presence of a modal marker. The semantic category of root modal is solely set as “volitional”, given that other categories mentioned in Cournane’s (2020) and Gibbs’ (1990) studies are incompatible with counterfactual construction. For instance, the semantic category of “ability” is encoded in *she can swim across the pool*; the semantic category of “permission” is encoded in *may I borrow your pencil*; the semantic category of “possibility” is encoded in *the car might go down the alley*. **Table 4** and **Table 5** show a sample set of stimuli with modal markers underlined. The participants will read separate versions of Chinese and English stimuli in random order with no content correspondence.

Table 4. A sample set of Chinese stimuli

condition	sample sentence
root modal (volitional)	如果/吉米/昨天/ <u>愿意</u> /交论文, /他/现在/就可以/休息。
chronological	If only Jimmy <u>would</u> have submitted the thesis yesterday, he could be

	having a rest now.
root modal (volitional) reverse chronological	如果/吉米/明天/ <u>愿意</u> /交论文，/他/现在/就不可以/休息。 If Jimmy would be submitting the thesis tomorrow, he could not be having a rest now.
no modal chronological	如果/吉米/昨天/交论文，/他/现在/就可以/休息。 If Jimmy had submitted the thesis yesterday, he could have a rest now.
no modal reverse chronological	如果/吉米/明天/交论文，/他/现在/就不可以/休息。 If Jimmy is going to submit the thesis tomorrow, he can't have a rest now.

Table 5. A sample set of English stimuli

condition	sample sentence
root modal (volitional) chronological	If only Jimmy <u>would</u> have revised his data yesterday, he could be finishing his analysis now.
root modal (volitional) reverse chronological	If Jimmy <u>would</u> be revising his data tomorrow, he could not be finishing his analysis now.
no modal chronological	If Jimmy had revised his data yesterday, he could finish his analysis now.
no modal reverse chronological	If Jimmy is going to revise his data tomorrow, he can't finish his analysis now.

A total of 480 critical sentences will be created, which will be assigned to 4 item sequences using a Latin square design to ensure that each condition per experimental set will appear only once in each sequence. Each participant shall read 120 critical sentences, counterbalanced by 30 counterfactual sentences in each condition. To reduce the saliency of the critical sentences, 60 filler sentences will be created. Holistically, critical sentences and fillers will be pseudorandomized so that the participants will not consecutively read more than two sentences with any identical condition. The participants will answer a simple follow-up question after each trial, which targets the antecedent clause of counterfactuals to verify comprehension (e.g. “Is it Samuel who was not willing to submit the thesis?”). Piloting will ensure clarity and balance (50% correct answers).

Procedure

The participants will first complete a consent form and a language background questionnaire to confirm their eligibility for the study. Upon arrival at the fMRI facility, they will receive a detailed briefing about the experimental tasks and safety protocols for MRI scanning. The experiment will consist of two sessions: a practice session outside the scanner and the main fMRI session.

In the practice session, the participants will familiarize themselves with the task by reading 20 sample sentences (10 Chinese and 10 English) presented on a computer screen. Each sentence will appear phrase-by-phrase, with each phrase displayed for 700ms followed by a 100ms blank screen, mimicking the pacing of the fMRI task. After each sentence, a comprehension question with yes/no options will assess their understanding. This session ensures participants are comfortable with the task format and minimizes learning effects during scanning.

For the fMRI session, the participants will lie supine in the scanner with their head stabilized using foam padding to minimize motion. Visual stimuli will be projected onto a screen at the rear of the scanner and viewed via a mirror attached to the head coil. The experiment will employ an event-related design with jittered inter-stimulus intervals (ISIs) optimized using OptSeq2 to maximize detection power. Each trial will begin with a fixation cross (500ms), followed by the sentence presented phrase-by-phrase (700ms per phrase, 100ms gaps). After the sentence, a comprehension question will appear for 4000 ms, requiring a button press (left for “yes” and right for “no”). Trials without questions will include a 4.8s fixation cross as a baseline. The task will be divided into four 10-minute runs, with short breaks between runs to prevent fatigue. Stimuli presentation will be controlled by using PsychoPy, and responses will be recorded via an MRI-compatible button box.

To counterbalance language and condition effects, 4 stimulus lists will be created by using a Latin square design, ensuring each participant sees each condition equally across languages. We acknowledge that comparing the same individuals processing two distinct languages may introduce broad differences in brain activation. To ensure that any observed contrast specifically reflects the cognitive processes of counterfactual reasoning rather than general language differences, we will implement two key controls. First, the order of language blocks (Chinese/English) will be fully randomized across the participants. Second, and crucially, language will be counted as a fixed

factor in all statistical models. This methodological approach can statistically isolate the effects of modality and proficiency from the general neural differences between L1 and L2 processing, thereby strengthening the inference that our findings pertain to counterfactual reasoning itself.

Before scanning, the participants will undergo a structural T1-weighted scan (5min) for anatomical localization, followed by functional EPI scans (TR=2000 ms, TE=30ms, flip angle=90°, 3mm isotropic voxels) covering the whole brain. Total scan time will be approximately 60 minutes, including setup and debriefing.

Data Analysis

Preprocessing of fMRI data will be conducted by using SPM12. Functional images will be realigned to correct for head motion, coregistered to the individual's T1-weighted anatomical scan, and normalized to the MNI template using DARTEL for improved intersubject alignment. Data will be smoothed with an 8 mm FWHM Gaussian kernel to account for anatomical variability and enhance signal-to-noise ratio. A high-pass filter (128s cutoff) will remove low-frequency drifts, and autoregressive modeling (AR(1)) will correct for temporal autocorrelation.

The first-level analysis will model each condition (modal/chronological, modal/reverse-chronological, non-modal/chronological, non-modal/reverse-chronological) as separate regressors convolved with the canonical hemodynamic response function (HRF). Comprehension questions and motion parameters will be included as nuisance regressors. Contrast images for each condition versus baseline (fixation) will be generated per participant.

Second-level random-effects analysis will employ a flexible factorial design with factors of language (Chinese, English), proficiency (intermediate, advanced), modality (modal, non-modal), and chronology (chronological, reverse-chronological). Main effects and interactions will be assessed by using F-tests, with follow-up t-tests to explore specific contrasts (e.g. modal>non-modal in L2). Whole-brain results will be thresholded at voxel-wise $p < 0.001$ (uncorrected) with cluster-level FWE correction at $p < 0.05$. Regions of interest (ROIs) in the left inferior frontal gyrus (IFG), temporoparietal junction (TPJ), and caudate nucleus---hypothesized to

be involved in modal and chronological processing---will be examined using small-volume correction with 10mm spheres centered on coordinates from meta-analyses.

The selection of IFG, TPJ, and caudate nucleus as core ROIs is theoretically grounded in their established roles in syntactic unification (IFG: Hagoort, 2013), mental-state reasoning for counterfactual scenarios (TPJ: Saxe & Kanwisher, 2003), and procedural learning of grammatical mappings (caudate: Ullman, 2001). Crucially, while default mode network regions may contribute to general hypothetical reasoning, the chosen ROIs specifically align with the linguistic demands of modal processing: IFG handles syntax-semantics integration of modal verbs (particularly for root modals requiring structural unification), TPJ supports perspective-shifting essential for hypothetical modality evaluation, and the caudate mediates the proceduralization of form-meaning associations that differ between L1 and L2 processing. This targeted circuit better captures the hierarchical complexity gradient from syntactic (root) to conceptual (complex) modal processing than domain-general networks.

Alongside the core language ROIs, we pre-register a complementary set of domain-general control ROIs. This control family includes the bilateral dorsolateral prefrontal cortex (BA 9/46), dorsal anterior cingulate cortex/pre-supplementary motor area (dACC/pre-SMA), and anterior insula/inferior frontal cortex (aINS/IF), regions robustly implicated in cognitive control and L2 processing. Furthermore, to adequately capture interhemispheric plasticity, we plan to include the right-hemisphere homologues of the left IFG and TPJ. All ROIs are defined a priori using spheres centered on peak coordinates from relevant meta-analyses. We will apply family-wise error correction (e.g. Holm-Bonferroni; see Wolkenhauer, 2013) across the two ROI families (“language” and “control”) to account for multiple comparisons. This comprehensive approach ensures that our analysis is sufficiently powered to detect effects in both canonical language networks and the domain-general control systems recruited during L2 processing.

The specific Montreal Neurological Institute (MNI) coordinates, sphere radii, and the corresponding meta-analytic or atlas sources for each ROI will be detailed in the supplementary materials. For inference, we will employ a two-tiered approach: whole-brain results will be assessed by using cluster-level family-wise error correction at $p < .05$, with a cluster-forming threshold of $p < .001$. For

our pre-registered ROIs, small-volume correction will be applied by using the pre-specified spheres. Effect sizes for significant contrasts within these ROIs will be reported.

Behavioral accuracy and reaction times for comprehension questions will be analyzed using mixed-effects models in R studio, with fixed effects for language, proficiency, modality, and chronology, and random intercepts for participants and items. Models will include interaction terms to test hypotheses about proficiency-modulated effects. Incorrect trials (>20% errors) will be excluded from fMRI analyses, and outliers ($RTs \pm 2.5$ SDs) will be trimmed.

All statistical maps will be visualized using xjView and reported according to COBIDAS guidelines, with cluster coordinates referenced to the AAL3 atlas. This analytical pipeline balances hypothesis-driven ROI approaches with exploratory whole-brain analyses to capture both predicted and emergent neural patterns.

Experiment 2

Participants

We plan to recruit an additional set of 30 neurotypical L1-Chinese L2-English learners with the same age range, educational background, and English proficiency profile as experiment 1. The exclusion criteria remain the same as in experiment 1.

Materials

Following experiment 1, we aim to investigate how different modal types embedded in counterfactual construction, including root and complex modals, manipulate L1 and L2 readers' cerebral reactions without considering reverse event chronology. In light of the challenges in constructing natural-sounding Chinese counterfactual sentences with epistemic modals, we have decided to exclude the epistemic modal condition from this experiment. While the original design aimed to compare root (volitional) and complex modals (hypothetical and epistemic) following Gibbs (1990) and Cournane's (2020) studies, pilot testing revealed that many Chinese epistemic modals (e.g. 应该/得 'should' or 敢 'dare') yielded semantically awkward or unnatural counterfactual constructions, unlike their English counterparts. For instance, structures like "if

Jimmy should have submitted the thesis yesterday, he could have a rest now” translate poorly into Chinese while maintaining counterfactual logic. Consequently, we focus solely on volitional (愿意 ‘would’) and hypothetical (能 ‘could’) modals, which align more naturally with Chinese syntax and preserve the intended counterfactual semantics. This adjustment ensures ecological validity while retaining the core contrast between root and complex modals.

The experimental stimuli include 3 single conditions: root modal (volitional), complex modal (hypothetical), and no modal. **Table 6** and **Table 7** show a sample set of stimuli with modal markers underlined. The participants will read separate versions of Chinese and English stimuli in random order with no content correspondence.

Table 6. A sample set of Chinese stimuli

condition	sample sentence
modal (volitional)	如果/吉米/昨天/ <u>愿意</u> /交论文，/他/现在/就可以/休息。 If only Jimmy would have submitted the thesis yesterday, he could be having a rest now.
modal (hypothetical)	如果/吉米/昨天/ <u>能</u> /交论文，/他/现在/就可以/休息。 If Jimmy could have submitted the thesis yesterday, he could be having a rest now.
no modal	如果/吉米/昨天/交论文，/他/现在/就可以/休息。 If Jimmy had submitted the thesis yesterday, he could have a rest now.

Table 7. A sample set of English stimuli

condition	sample sentence
modal (volitional)	If only Jimmy <u>would</u> have revised his data yesterday, he could be finishing his analysis now.
modal (hypothetical)	If Jimmy <u>could</u> have revised his data yesterday, he could be finishing his analysis now.
no modal	If Jimmy had revised his data yesterday, he could finish his analysis now.

Similar to experiment 1, to ensure that each participant will read 120 critical sentences counterbalanced by 40 counterfactual sentences in each condition, a total of 360 critical sentences

will be created by using a Latin square design. 60 filler sentences will be created. Critical sentences and fillers will be pseudo-randomized.

Procedure

Following the same screening and consent procedures as in experiment 1, eligible participants will proceed directly to the fMRI session. Unlike experiment 1, which investigated interactions between modal markers and event chronology, experiment 2 focuses exclusively on how different modal types (root VS complex) modulate neural responses in L1 and L2 processing.

Stimulus presentation will mirror the phrase-by-phrase design in experiment 1 (700ms/phrase, 100ms gaps), but critical sentences will now include 4 modal conditions. The phrase-by-phrase presentation rate was determined through a rigorous pilot test to ensure it was sufficient for accurate comprehension across all conditions and did not impose a differential processing burden. To further mitigate the potential confound of working memory load, the total number of phrases per sentence will be included as a covariate in the behavioral analysis. For the fMRI design, the inter-stimulus interval will be jittered by using OptSeq2, with the optimal inter-stimulus interval distribution determined by the software to maximize design efficiency for estimating the hemodynamic response. Trials without comprehension questions will include a 4.8s fixation cross, which serves as an explicit baseline for the event-related design.

To minimize fatigue, the experiment will be split into four 10-minute runs, with randomized language blocks (Chinese/English) and counterbalanced condition orders across participants. Each run will contain 30 critical sentences (7-8 per modal condition) interspersed with 15 filler sentences, pseudo-randomized to avoid consecutive repetitions of the same condition.

A key modification is the inclusion of a post-scan behavioral debriefing, where participants will complete a subset of the comprehension questions again outside the scanner. This will assess whether offline accuracy correlates with neural activation patterns, particularly for hypothetical modals, which are hypothesized to impose higher cognitive demands on L2 learners.

Data Analysis

Pre-specified data quality checks include verifying that comprehension question accuracy (behavioral data) does not exhibit ceiling effects (>90% correct) or floor effects (<70% correct) across conditions. fMRI preprocessing will include visual inspection of EPI data for artifacts.

Preprocessing pipelines (realignment, coregistration, normalization, smoothing) will replicate Experiment 1, but first-level models will now incorporate 3 modal-specific regressors (root, hypothetical, no-modal) convolved with the canonical HRF. Nuisance regressors will include comprehension question trials and motion parameters, but outlier trials (e.g. incorrect responses or excessive head motion) will be flagged and excluded at this stage.

At the second level, analyses will focus on three primary investigative directions to address the core hypotheses of experiment 2. The first analytical strand will examine a modal complexity gradient, testing whether neural activation follows a hierarchical pattern of increasing intensity corresponding to semantic complexity levels (non-modal<root<hypothetical), with particular attention to regions implicated in syntactic unification (left IFG) and mental-state reasoning (TPJ). This approach will reveal whether the brain distinguishes between modal categories based upon their conceptual demands. The second analytical dimension will explore proficiency modulation effects by incorporating continuous proficiency scores from standardized tests as parametric modulators, allowing the identification of brain regions where L2 learners' neural responses systematically vary with language competency, especially crucial for processing high-complexity hypothetical modals where proficiency effects are predicted to be most pronounced. The third investigative axis will directly compare whole-brain activation patterns between native Chinese and L2-English groups, with ROI analyses concentrating on the caudate nucleus (associated with procedural learning of grammatical structures) and dorsolateral prefrontal cortex (involved in cognitive control mechanisms), areas where L1/L2 processing differences are theoretically expected to emerge most clearly. These three analytical approaches will be complemented by functional connectivity analyses examining dynamic interactions between key regions during modal processing.

Functional connectivity (PPI) analyses will examine the dynamic coupling between the left IFG and TPJ during hypothetical modal processing, probing whether L2 learners rely more heavily on top-down control mechanisms. To address multiple comparisons, cluster-level FWE correction

($p < 0.05$) will be applied, with small-volume corrections for ROIs.

Exploratory analyses will leverage multivariate pattern analysis (MVPA) to decode modal-type representations from spatially distributed activation patterns, offering finer-grained insights into how L1 and L2 neural representations diverge. All code and thresholded statistical maps will be made publicly available to ensure reproducibility.

Finally, we anticipate completing the entire study within 2 months of approval, with stage 2 submission following 1 month after data analysis. The experimental stimuli, preprocessing and first-level analysis code, pre-registered exclusion criteria, and thresholded statistical maps will be made publicly available upon publication. All exploratory analyses (e.g. MVPA, jackknife) will be clearly distinguished from the pre-registered confirmatory analysis plan. Following the confirmatory analyses, exploratory analyses including parametric modulation of proficiency, jackknife robustness checks of cluster-level results, and psychophysiological interaction will be conducted.

Data Availability

Data will be available if the stage 1 manuscript is in principle accepted.

Human Ethics Statements

The study received informed consent from all participants, and no confidential information was collected.

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